

# TARGET-90

**JKBOSE CLASS-11<sup>TH</sup>**

## LAW'S OF MOTION

One Shot

**PHYSICS**

**By-Wajahid Sir (Gabbar Sir)**





## Topics to be Covered

- A ... laws of motion (I, 2, 3)
- B ... Types of force (friction)
- C ... Banking of Road (3 case)
- D ... Centripetal Force

## Unit III: Laws of Motion (07 marks)

### Laws of Motion

Intuitive concept of force, Inertia, Newton's first law of motion; momentum and Newton's second law of motion; impulse; Newton's third law of motion. Law of conservation of linear momentum and its applications. Equilibrium of concurrent forces, Static and kinetic friction, laws of friction, rolling friction, lubrication. Dynamics of uniform circular motion: Centripetal force, examples of circular motion (vehicle on a level circular road, vehicle on a banked road).





# Learning Objectives



**Inertia**



**Newton's 1<sup>st</sup> Law of Motion**



**Momentum**



**Newton's 2<sup>nd</sup> Law of Motion**



Think  
Learn  
Grow



# Learning Objectives



**Impulse**



**Newton's 3<sup>rd</sup> Law of Motion**



**Law of Conservation of momentum**



**Recoil of Gun**



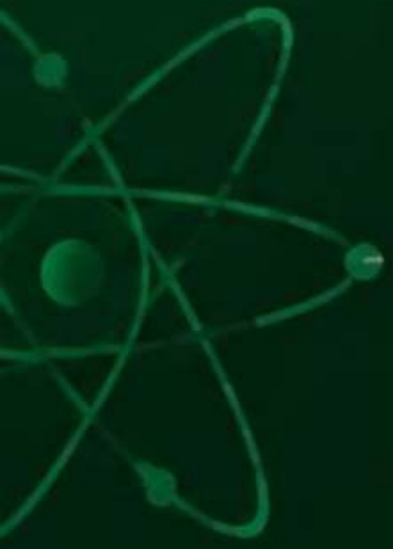
Think  
Learn  
Grow



# Learning Objectives



$$F=ma$$



**Angle of Repose**



**Centripetal Force**



**Maximum safe velocity of a car on a level road**



**Maximum safe velocity of a car on a banked road**

Think  
Learn  
Grow



# Learning Objectives



**Rocket Propulsion**



**Equilibrium of Force**



**Friction**



**Graphs**



$$F=ma$$



Think  
Learn  
Grow



## Inertia

↓  
Susti

Inertia  $\propto$  Mass

- (•) Inertia is the property of Body by virtue  
It does not change its State of Motion ✓  
Until and unless any External  
force is applied.
- Rest ✓  
Direction ✓



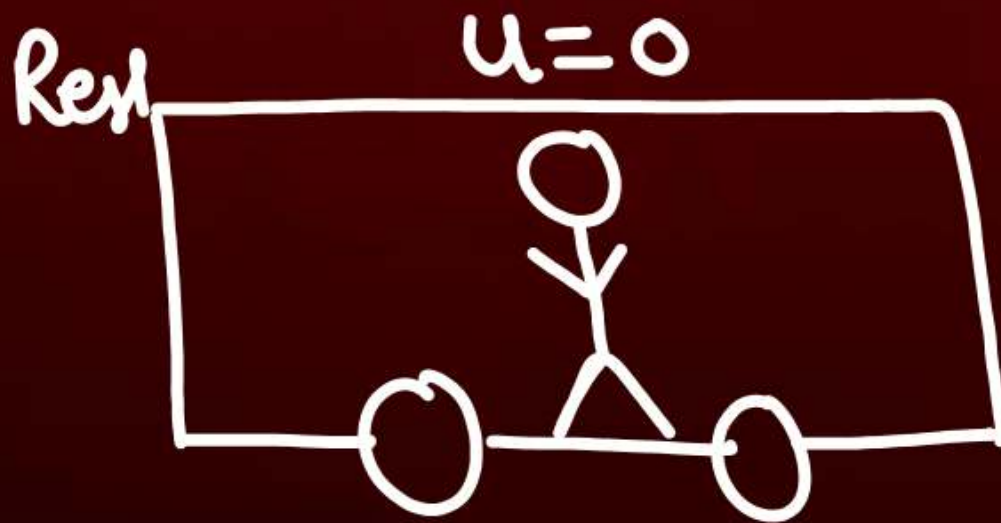
# Inertia

## Type of inertia

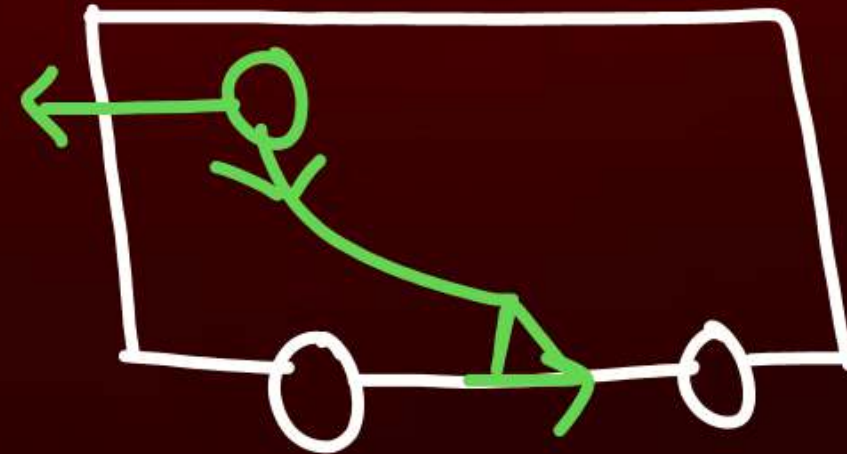
3 type

### 1. Inertia of Rest

The properties of the object by which it will remain in rest until and unless any external force is applied.



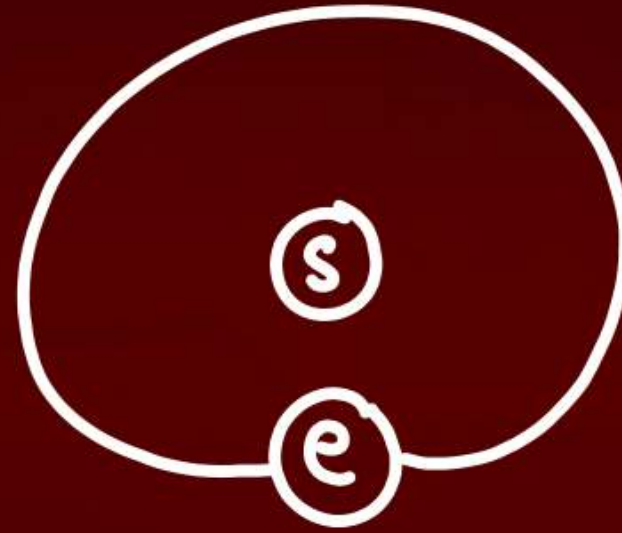
Backward





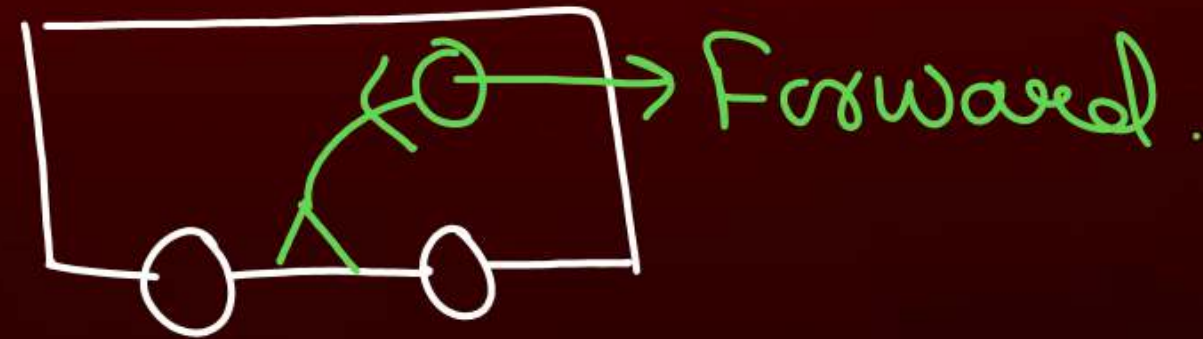
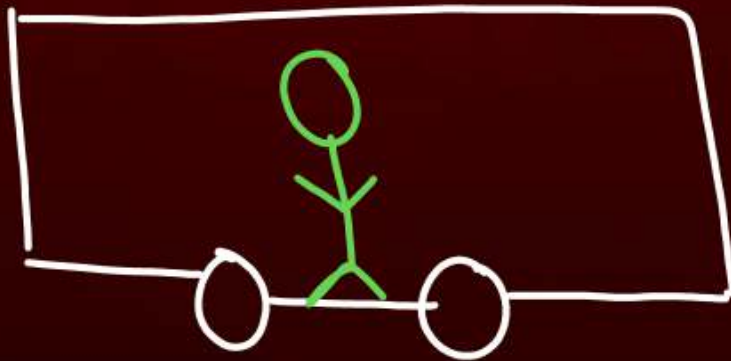
# Inertia

## 2. Inertia of Motion



The properties of the object by which it will remain in **motion** until and unless any external force is applied. ✓

$$V = 2 \text{ m/s}$$



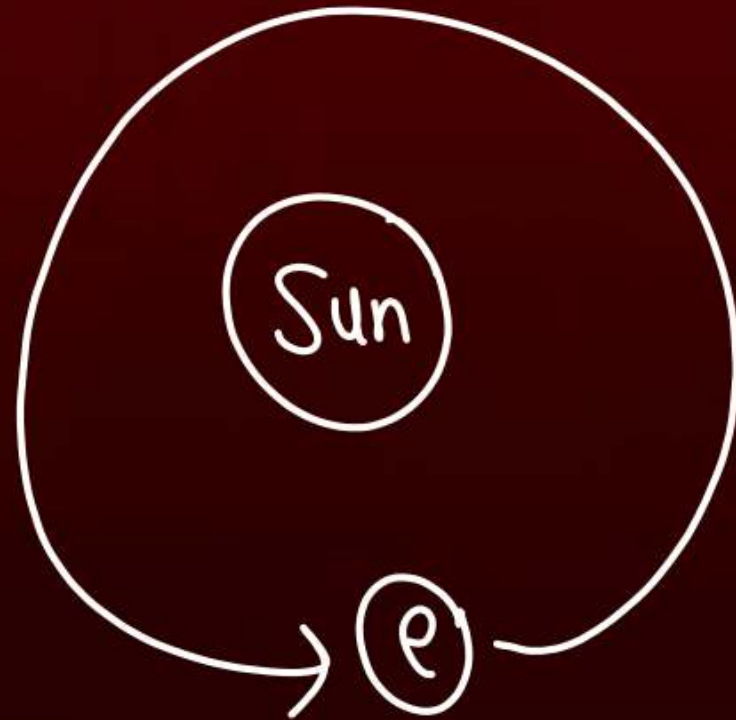


# Inertia

## 3. Inertia of Direction

Particular

The properties of the object by which it will remain in **direction** until and unless any external force is applied.





## Newton's 1<sup>st</sup> Law of Motion

→ Explain the Inertia



Properties of the object by which it will remain in rest motion / any particular direction until and unless any external force is applied.

Newton 1st law



law of Inertia





# Momentum

$\vec{p}$  /  $\vec{P}$

- ❖ It give the quantity of motion.
- ❖ It is the product of mass and velocity.
- ❖ It is a vector.

$$\vec{P} = m\vec{v}$$

$$\text{Unit} = \text{kg m/s}$$

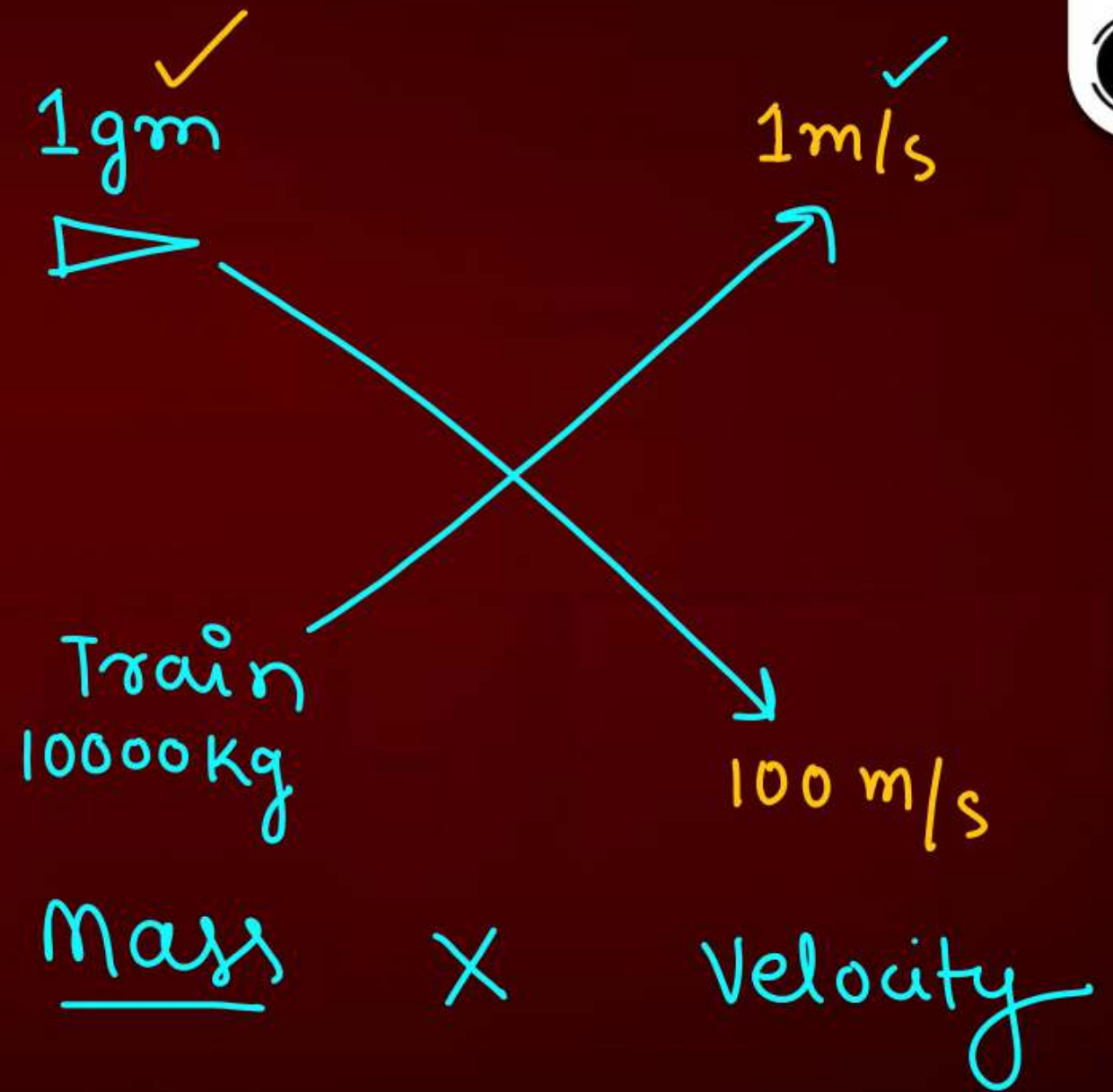
$$\vec{P}_f = mv$$

$$P_i = mu$$

$$\Delta P = P_f - P_i \\ = mv - mu.$$

$$\Delta P =$$

$$\Delta P =$$



$$p = \underline{\text{mass}} \times \text{velocity}$$

$$\uparrow\uparrow p = m\uparrow v\uparrow$$



# Newton's 2<sup>nd</sup> Law of Motion

# define force

The rate of change of momentum is called force.

$$F = \frac{\Delta p}{\Delta t}$$

$\Delta \rightarrow$  Bada change  
 $d \rightarrow$  chota change

$$F = \frac{dp}{dt}$$

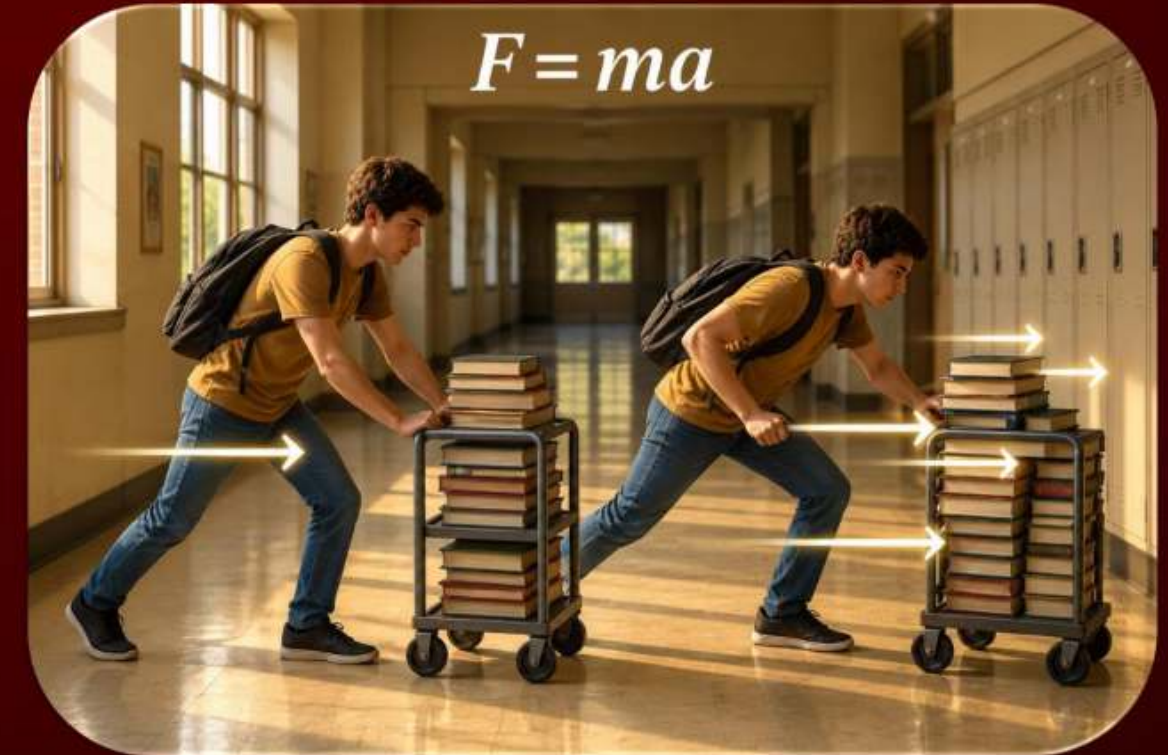
$$p = mv$$

$$F = \frac{d(mv)}{dt}$$

$$\left( \frac{d}{dt} uv = u \frac{dv}{dt} + v \frac{du}{dt} \right)$$

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

differential equation of force



$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

$$dm = 0$$



Special case

$$\Delta m = 0$$

$$dm = 0$$

$$F = m \frac{dv}{dt} + v \frac{dm}{dt} \quad \begin{matrix} m_f - m_i = 0 \\ m_f = m_i \end{matrix}$$

$$F = m \frac{dv}{dt} + v \times 0$$

$$F = m \frac{dv}{dt} + 0 \quad a = \frac{dv}{dt}$$

$$F = ma$$



## Newton's 2<sup>nd</sup> Law of Motion

$$F = ma$$

"Force is the product of mass and acceleration"



# Newton's 2<sup>nd</sup> Law of Motion

## Unit of Force

|   |                                            |                                               |
|---|--------------------------------------------|-----------------------------------------------|
| ❖ | $F = ma$                                   | $F = ma$                                      |
| ❖ | $F = \text{Kg m/s}^2$                      | $F = \text{gm } \frac{\text{cm}}{\text{s}^2}$ |
| ❖ | $F = \text{Newton}$                        | $F = \text{gm } \frac{\text{cm}}{\text{s}^2}$ |
| ❖ | $\underbrace{\hspace{1.5cm}}_{\text{S.I}}$ | $F = \text{dyne}$                             |

$$1 \text{ N} = 10^5 \text{ dynes.}$$



## Newton's 2<sup>nd</sup> Law of Motion

### What is 1 Newton Force?

- ❖  $F = ma$
- ❖  $F = 1 \text{ kg } 1 \text{ m/s}^2$
- ❖  $F = 1 \text{ kg m/s}^2$
- ❖  $F = 1 \text{ New}$

• The force which produce the  $a = 1 \text{ m/s}^2$  in body of 1 kg mass.





# Impulse

The force acting for a short period of time.

# 2nd law of motion

$$F = \frac{dp}{dt}$$

$$\uparrow F = \frac{dp}{dt} \downarrow \downarrow$$

$$\text{Impulse} = \underline{F \times dt}$$

$$\text{Impulse} = F dt = dp$$

Impulse momentum  
Theorem

Example:

Punch on a Boxing Bag: A boxer's fist hits a punching bag, applying a sudden force that changes the bag's motion.

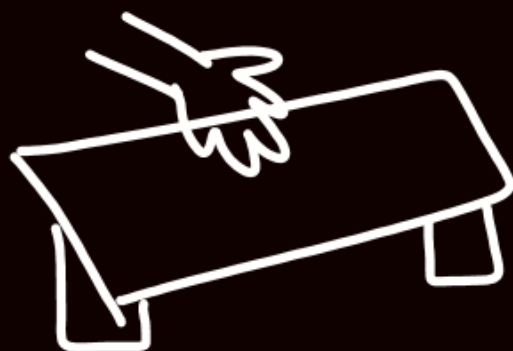


# Application of Impulse

$$F = \frac{dp}{dt}$$

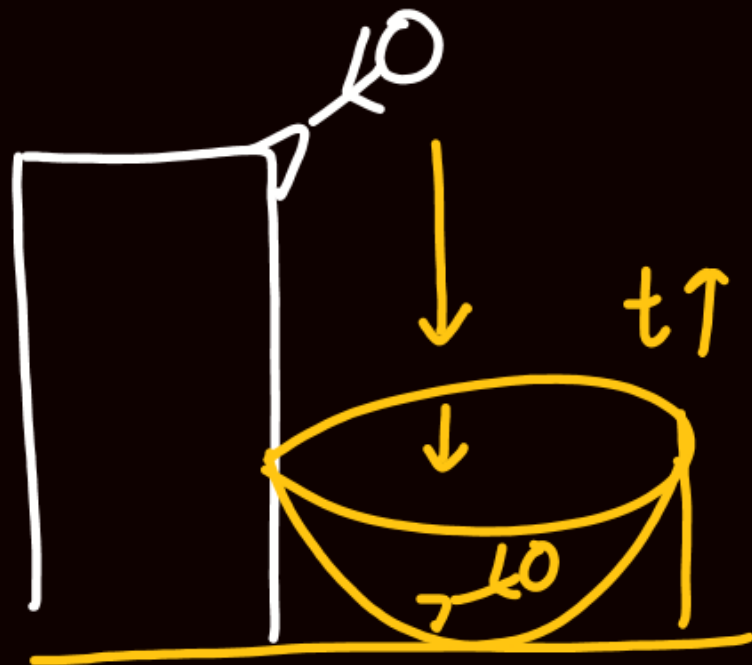


(1) Karate player



$t \downarrow F \uparrow$

(2)



$t \uparrow F \downarrow$

(3) Catching player



time  $\uparrow$   
Force  $\downarrow$



## Newton's 3<sup>rd</sup> Law of Motion

"For every action there is equal and opposite reaction."



Flying of Birds

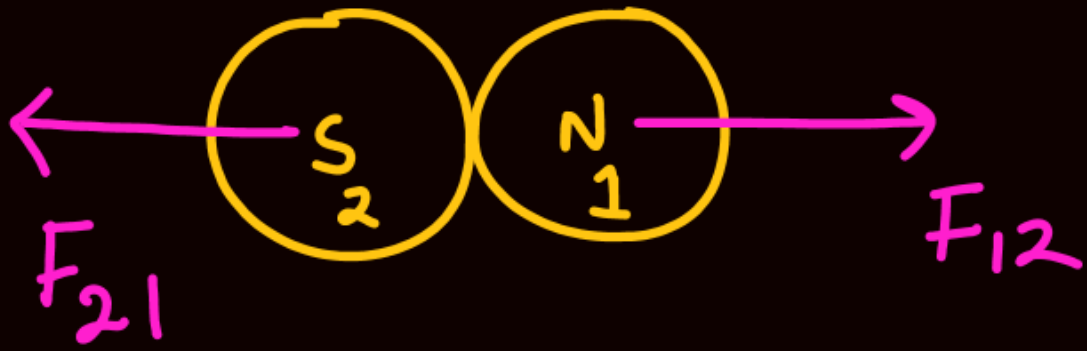


Recoil of gun



Swimming

# Newton 3rd law of Motion



Every Action there  
is equal and opposite  
Reaction

$$\text{Action} = -\text{Reaction}$$

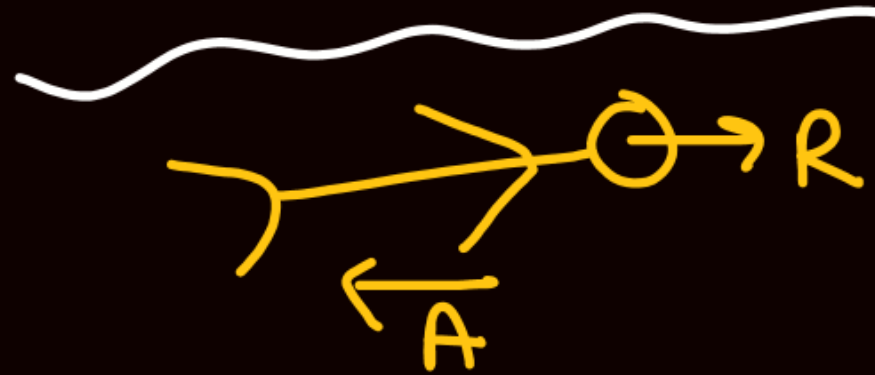
$$F_{12} = -F_{21}$$

- ① Force occur in pair
- ② Both force occur simultaneously
- ③ Both the force have same nature

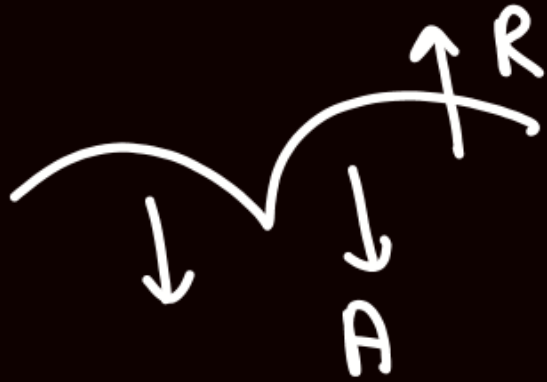
## (•) Application of Newton 3rd law of motion



(3) Swimming



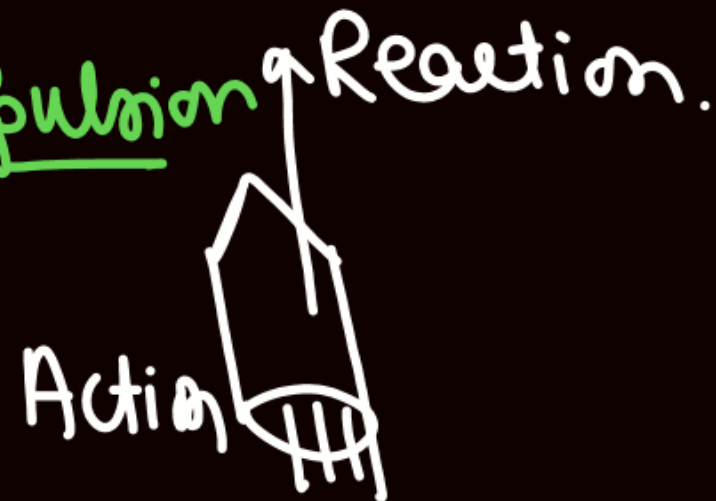
(2) Flying of Bird



(4) Recoiling of gun



(5) Rocket propulsion





# $P_i = P_f$ Law of Conservation of Momentum

If any external force is equal to zero then initial momentum is equal to the final momentum. ✓

(i) for one Body system

$$\textcircled{m} \rightarrow u$$

$$F = \frac{dp}{dt}$$

$$0 = \frac{dp}{dt}$$

$$\textcircled{m} \rightarrow u$$

$$0 \times dt = dp$$

$$0 = dp$$

$$0 = \Delta p$$

$$\underline{\Delta p} = 0$$

$$p_f - p_i = 0$$

$$p_f = p_i$$

$$mu = mu$$

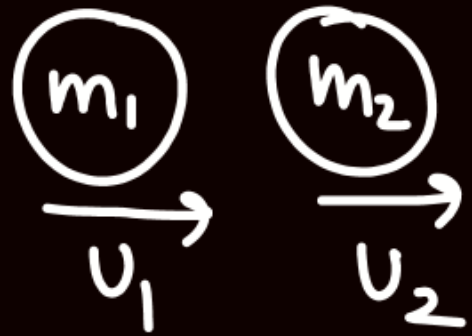
$P_i$

## (ii) For two Body System

$$F = \frac{\Delta p}{t}$$

let  $m_1$  and  $m_2$  are  
two masses moving  
with velocity  $u_1$  and  $u_2$

$$u_1 > u_2$$

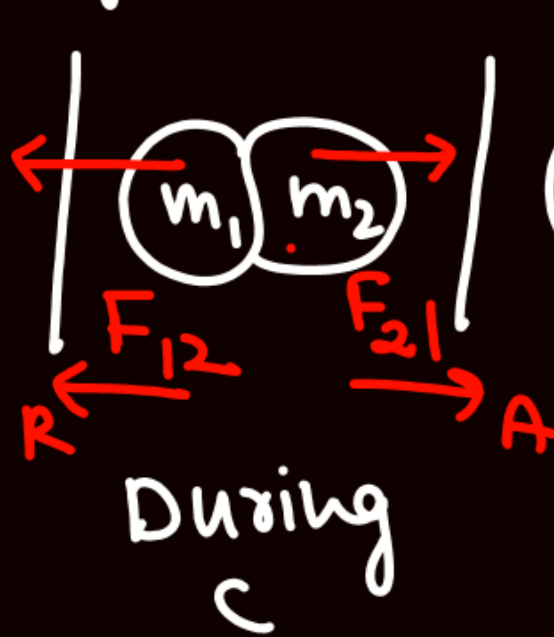


Before C.

initial

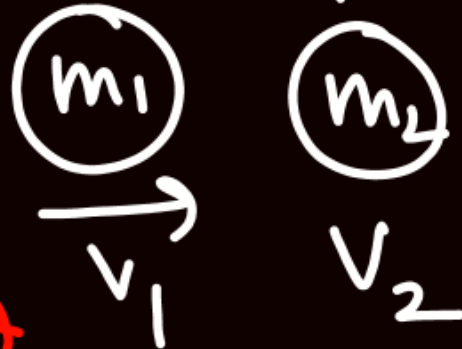
$$p_{i1} = m_1 u_1$$

$$p_{i2} = m_2 u_2$$



During  
C

$$v_2 > v_1$$



After collision

final

$$p_{f1} = m_1 v_1$$

$$p_{f2} = m_2 v_2$$

$$F_{21} = -F_{12}$$

$$\frac{\Delta p_2}{t} = -\frac{\Delta p_1}{t}$$

$$x = y$$
$$y = x$$

$$p_{f2} - p_{i2} = - (p_{f1} - p_{i1})$$

$$p_{f2} - p_{i2} = -p_{f1} + p_{i1}$$

$$p_{f2} + p_{f1} = p_{i1} + p_{i2}$$

$$m_2 v_2 + m_1 v_1 = m_1 u_1 + m_2 u_2$$

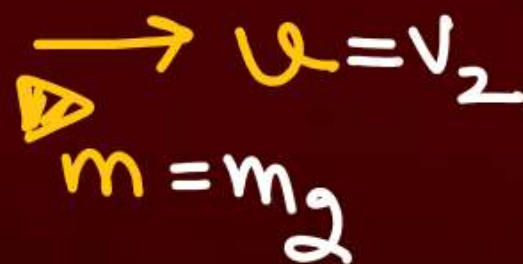
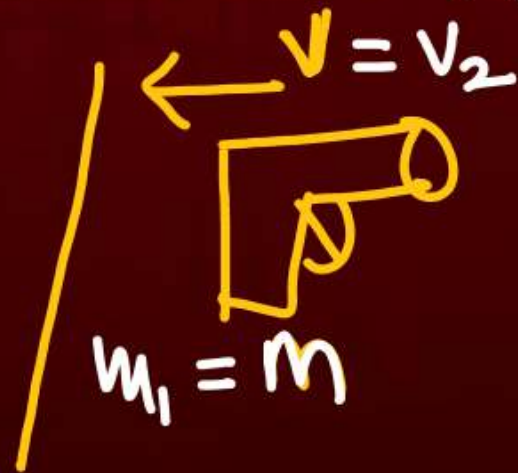
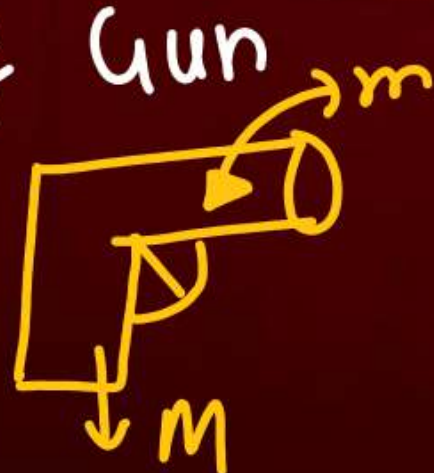
$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$



## Recoil of Gun

- When shoot the Gun the Bullet moves forward and Gun move Backwarded This is called Recoiling of Gun

$$\begin{aligned} m_1 &= M \\ m_2 &= m \\ u_1 &= 0 \\ u_2 &= 0 \end{aligned}$$



$$\underline{m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2}$$

$$M \times 0 + m \times 0 = M V + m u$$

$$0 + 0 = M V + m u$$

$$0 = M V + m u$$



$$0 = \underline{mV} + \underline{mv}$$

$$-mv = mV$$

$$-\frac{mv}{m} = V$$

$$\downarrow V = -\frac{mv}{m} \uparrow$$

Recoiling  
velocity of gun



## Rocket Propulsion

Application of Newton 3rd law of motion -

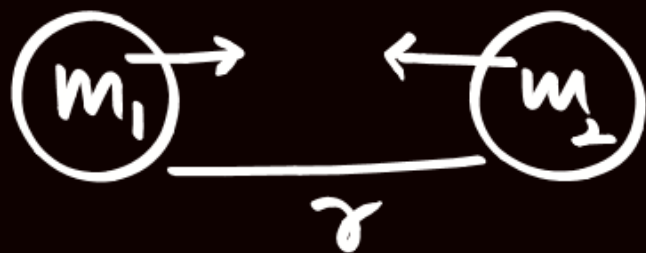
- ❖ you can observe that the mass of Rocket get decrease with time

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$



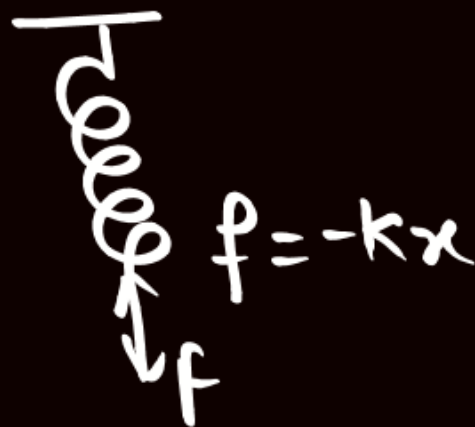
# Types of Force

(a) Gravitation force

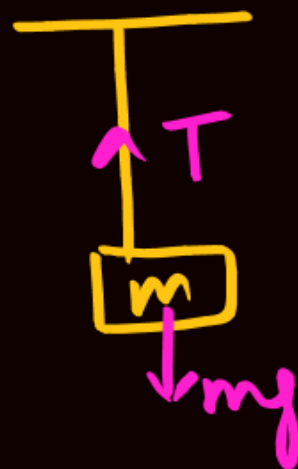


$$F = G \frac{m_1 m_2}{r^2}$$

(b) Spring force



(c) Tension



(d) Electrostatic force

+q      -q

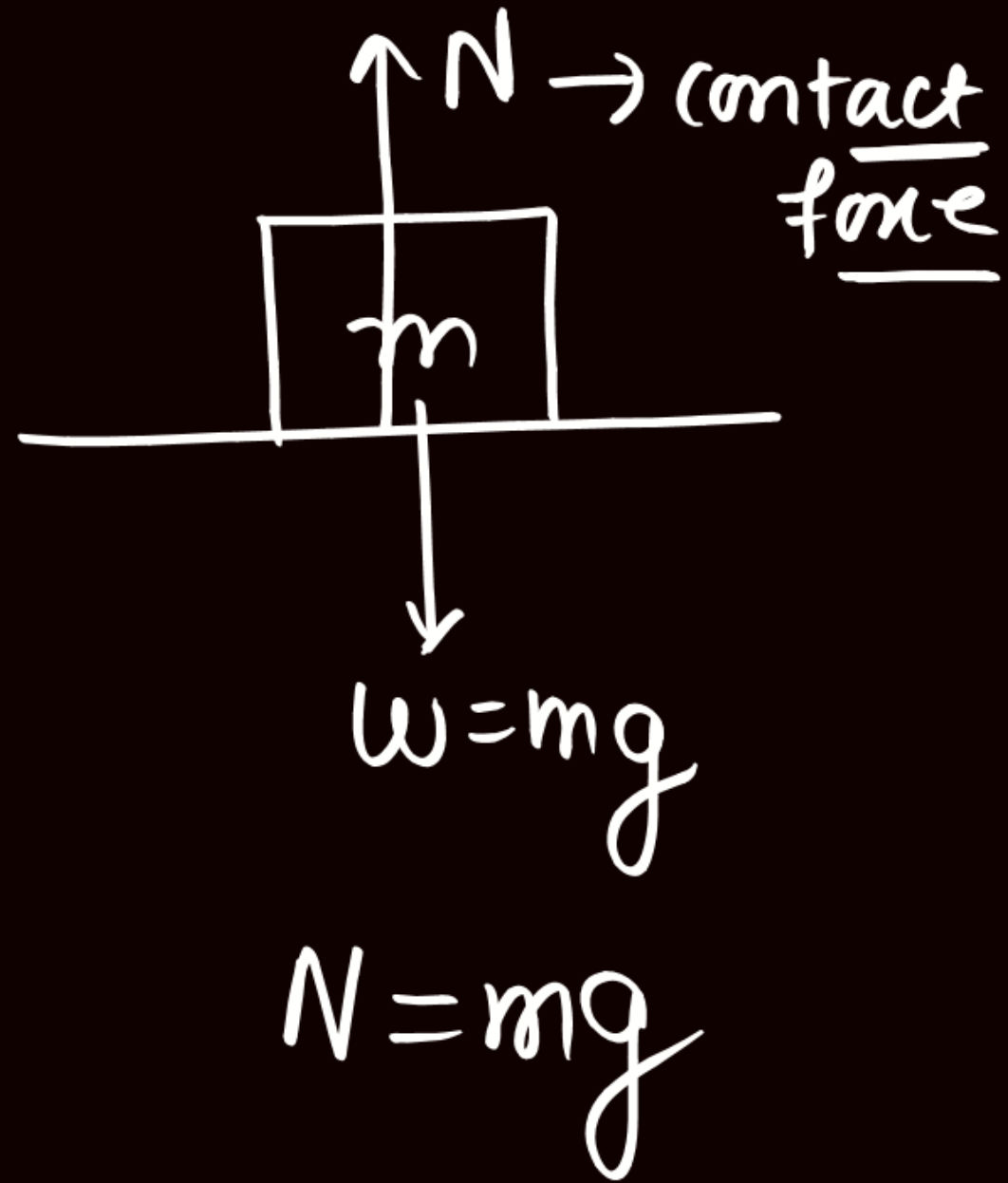
(e) friction force (detail)



(f) magnetic force



(g) Normal force





# Friction

The force which oppose the relative motion.

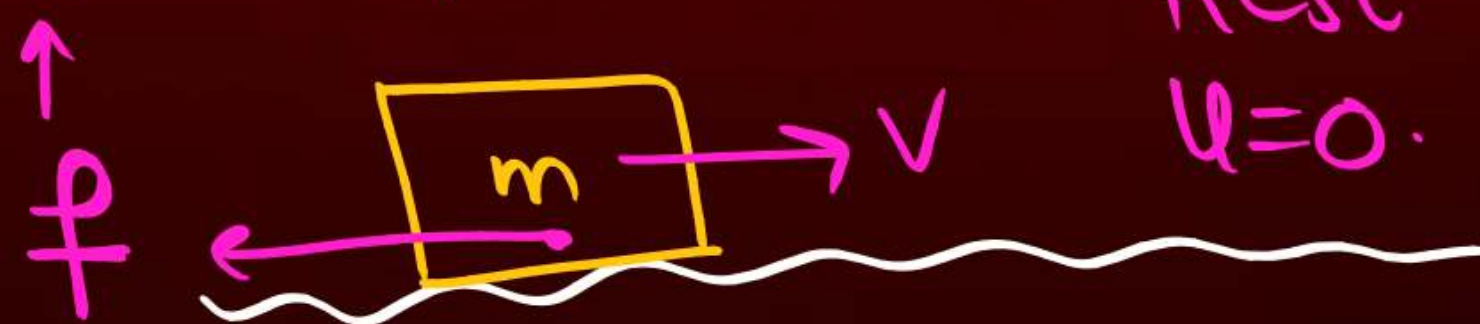
## Causes of Friction

### 1. Roughness

Roughness  $\uparrow\uparrow$   $f \uparrow$

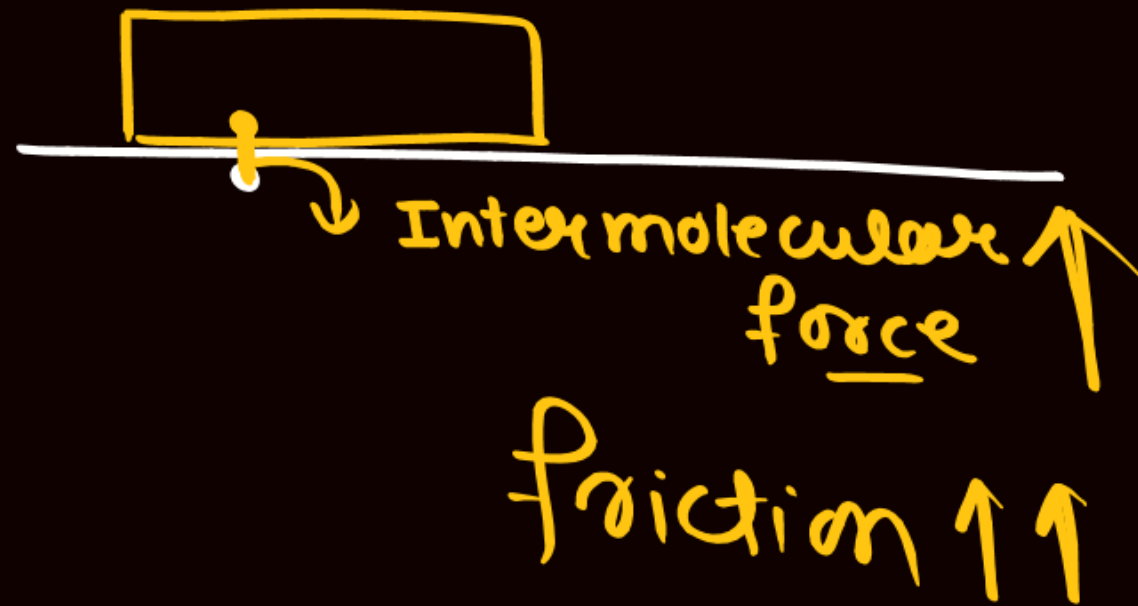
Smoothness  $\uparrow\uparrow$   $R \downarrow$   $f \downarrow$

friction force.



Rest  
 $u=0$ .

## (2) Extra Smoothness



# Types of friction

(a) Static friction ( $f_s$ )

the force of friction which act when Body is in Rest

# Static friction value  $\rightarrow$  change

# intelligent force  
 $f_s = \mu_s N$

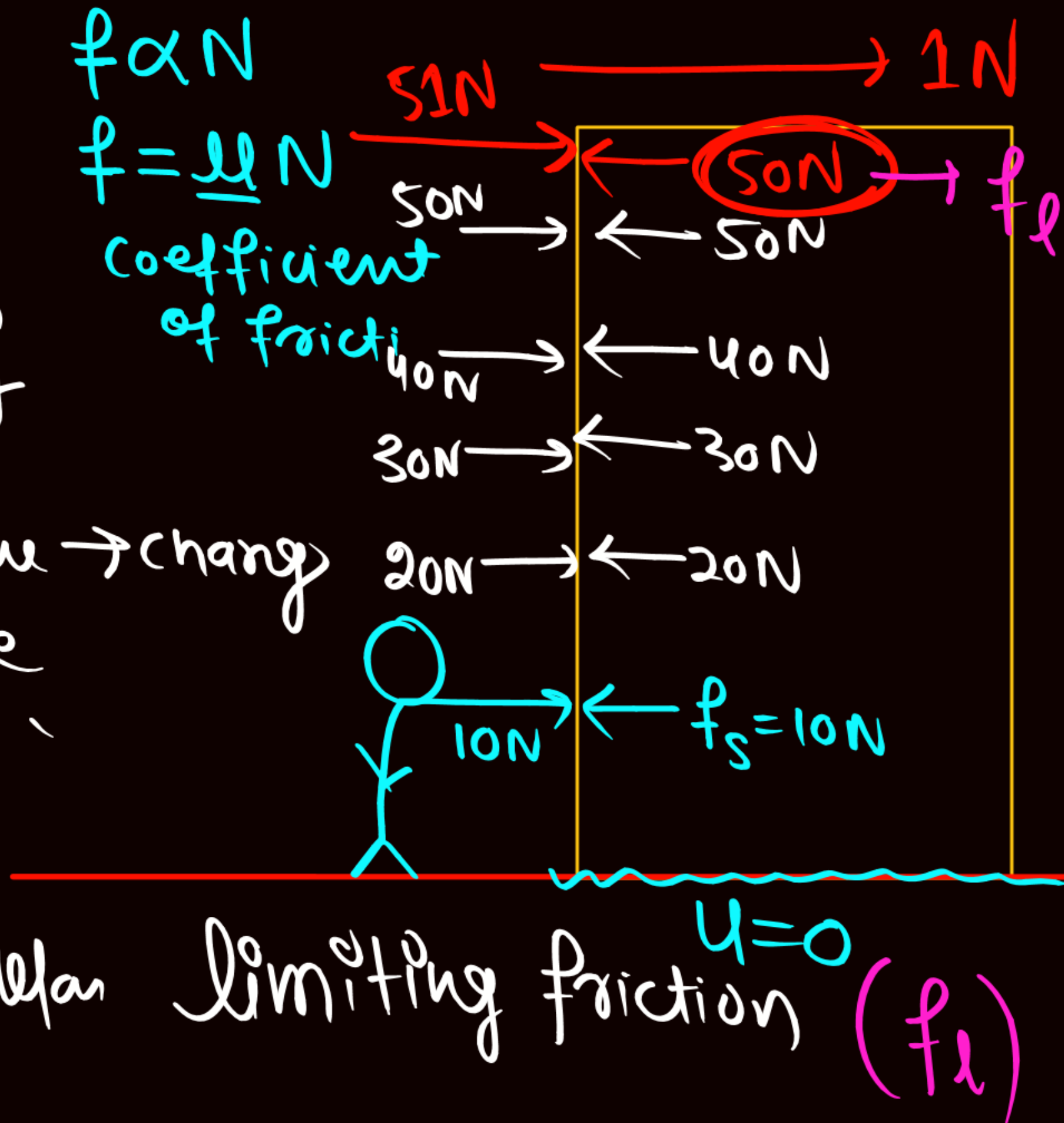
The maximum value

of Static friction is  $\mu_s N$

$$f \propto N$$

$$f = \mu N$$

coefficient  
of friction

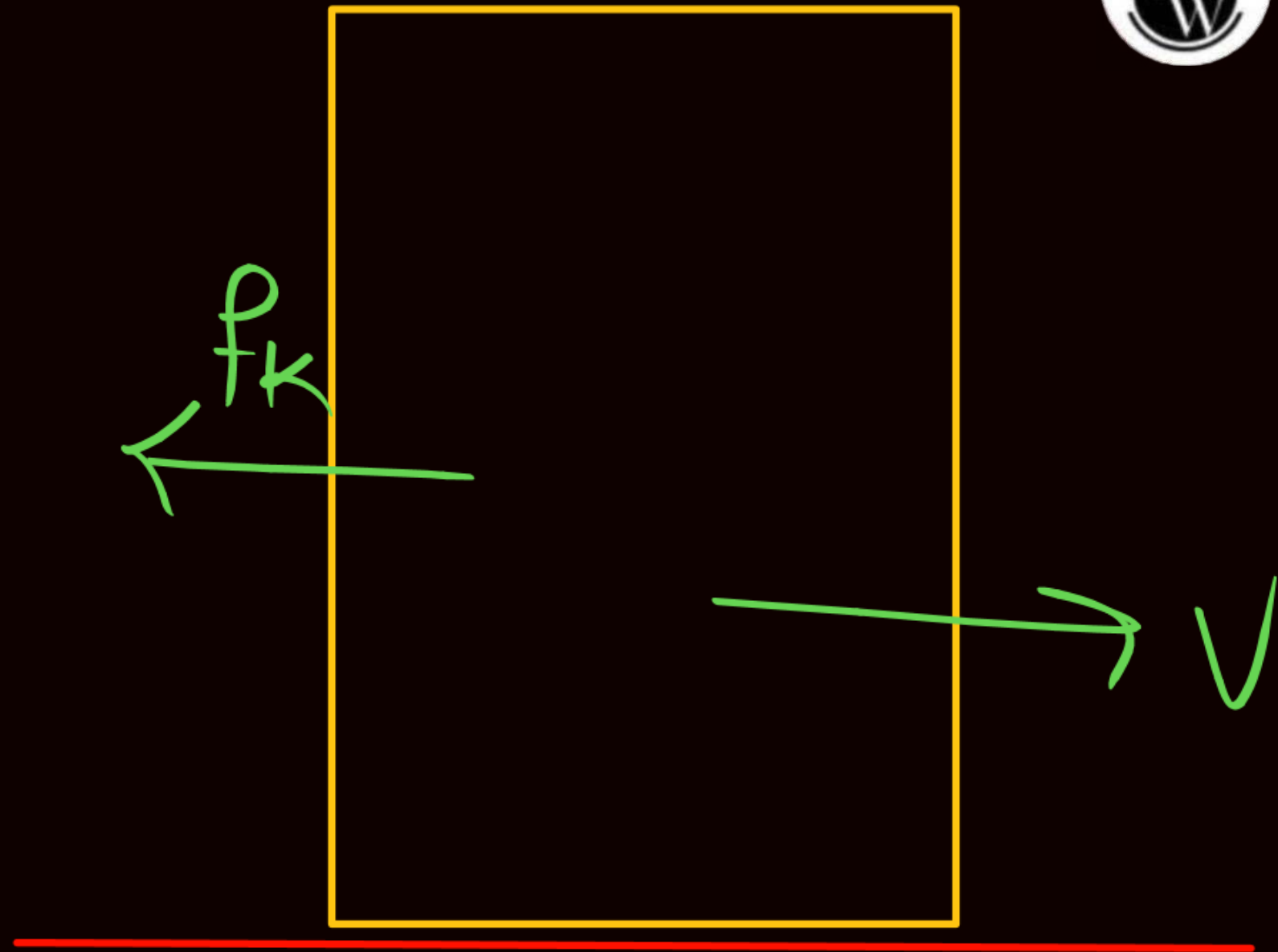


## (II) Kinetic friction ( $f_k$ )

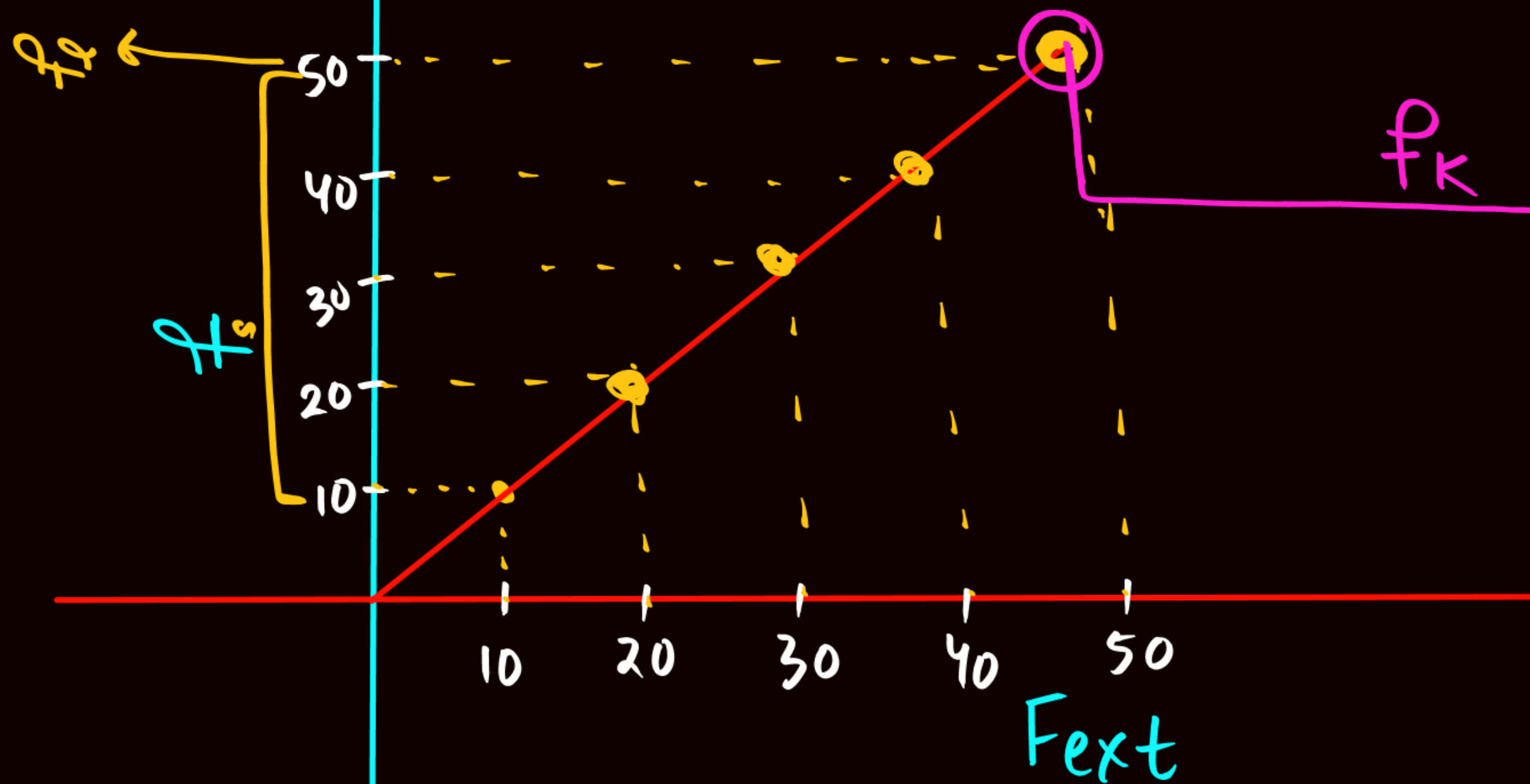
the friction force  
act when Body  
Just Start

$$f_k = \mu_k N$$

$$f_s < f_s$$



# friction graph





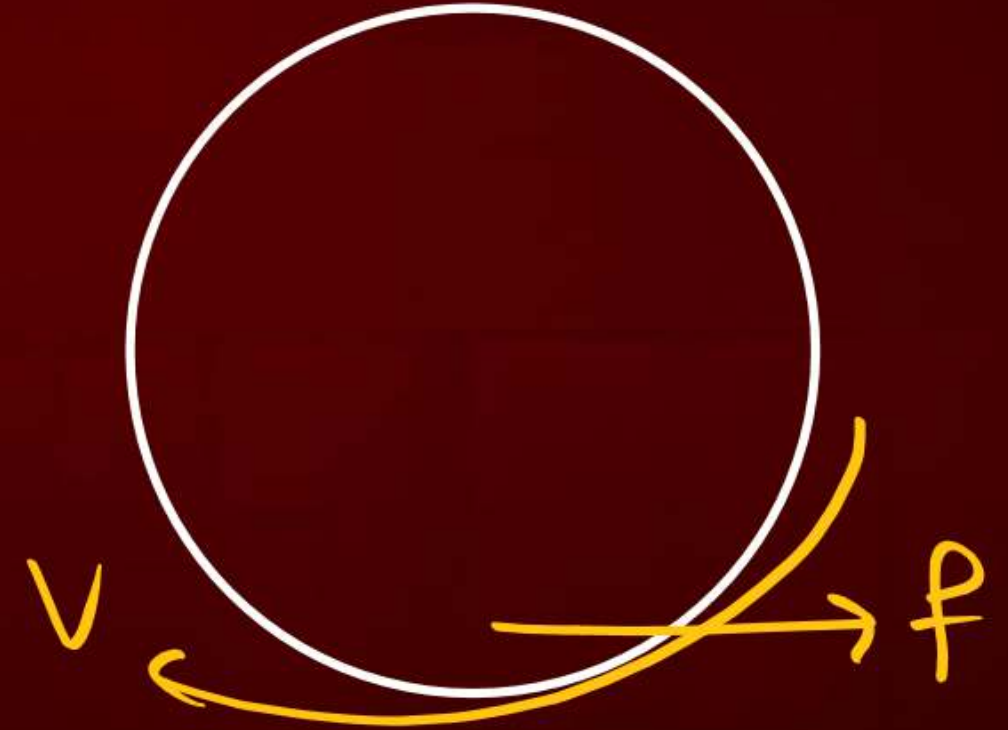
# Friction

## Types of Friction

### 3. Rolling Friction

The friction force which act when body rolls.

$$* * \boxed{f_r = \mu_r N}$$





# Friction

$$f \propto N$$

## Law's of Friction

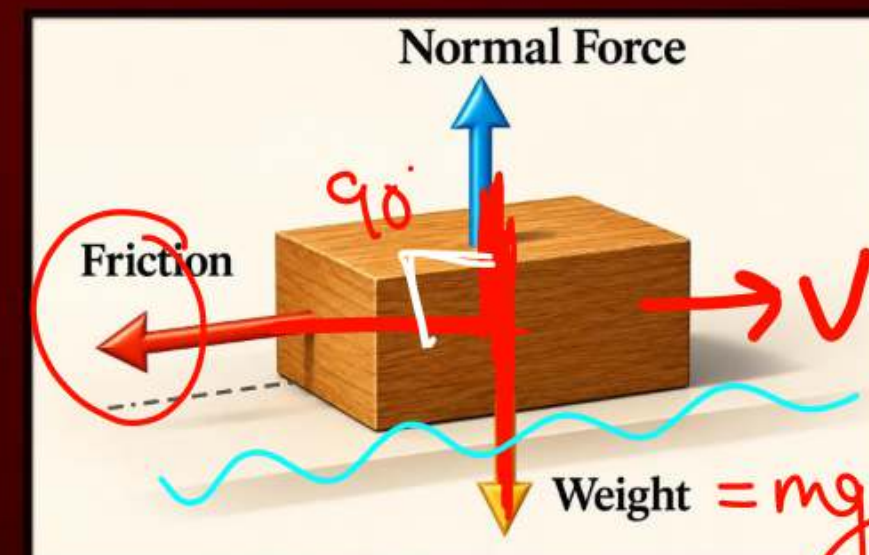
1 ❖

The friction of the moving object is proportional and perpendicular to the normal force.

❖

The friction experienced by the object is dependent on the nature of the surface it is in contact with.

$$N = mg$$



$$f \propto N$$

$$f = \mu N$$

↳ coefficient of friction



# Friction

$$f \propto N$$

$$f = \mu N$$

$$f_s = \mu_s N \quad f_k = \mu_k N$$

## Law's of Friction

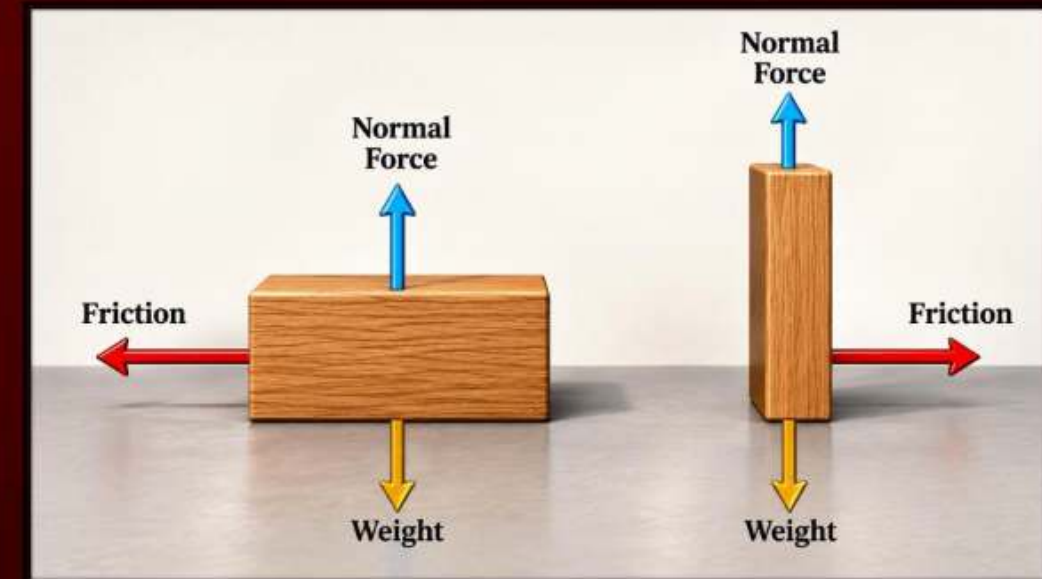
3 ✖

The coefficient of static friction is greater than the coefficient of kinetic friction.

$$f_s > f_k$$

$$\mu_s > \mu_k$$

$$\mu_s > \mu_k$$





# Friction

## Law's of Friction

$$\mu_S > \mu_K$$

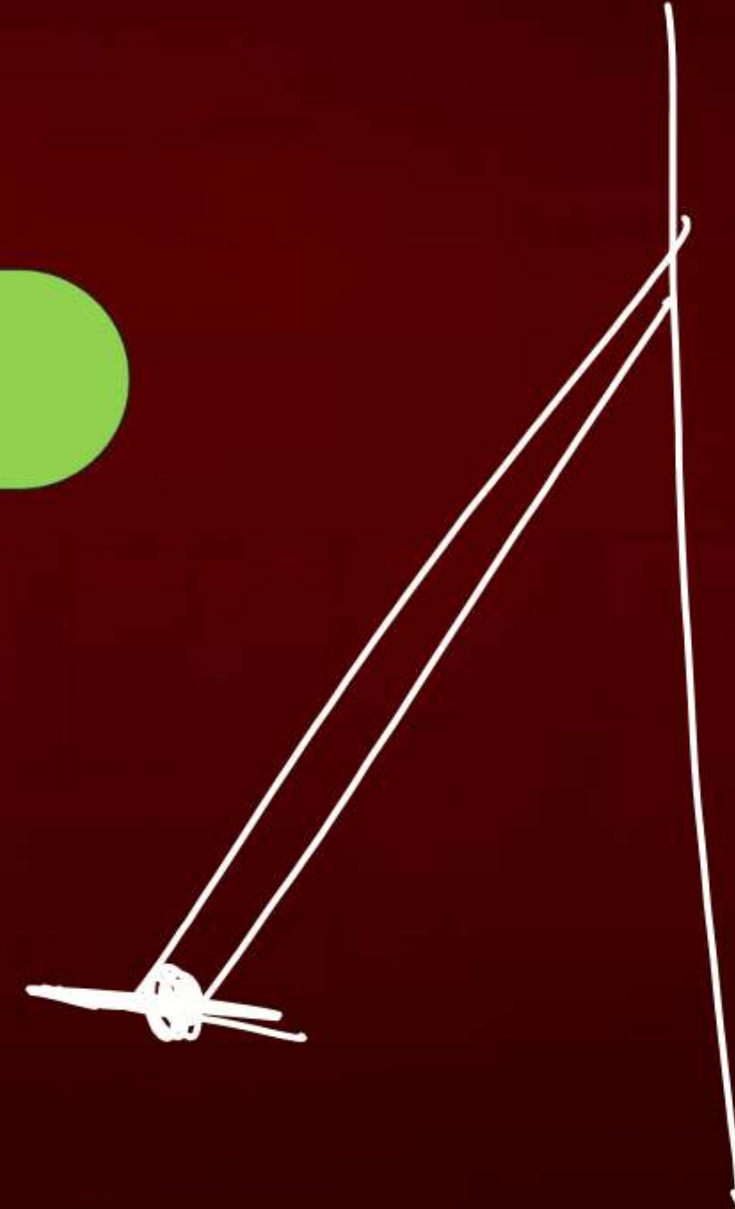




# Friction

## Advantages of Friction

- ❖ Friction enables us to walk freely. ✓
- ❖ It helps to support ladder against wall.
- ❖ Brakes of vehicles work due to friction.



# Waiting



# Friction

## Disadvantages of Friction

- ❖ Friction produces heat which damages the moving parts of a machine.
- ❖ Friction produces wear and tear on the contacting surfaces. this reduces the life of machine parts, tyres and shoe soles.
- ❖ A lot of energy is wasted due to friction to overcome it before moving.

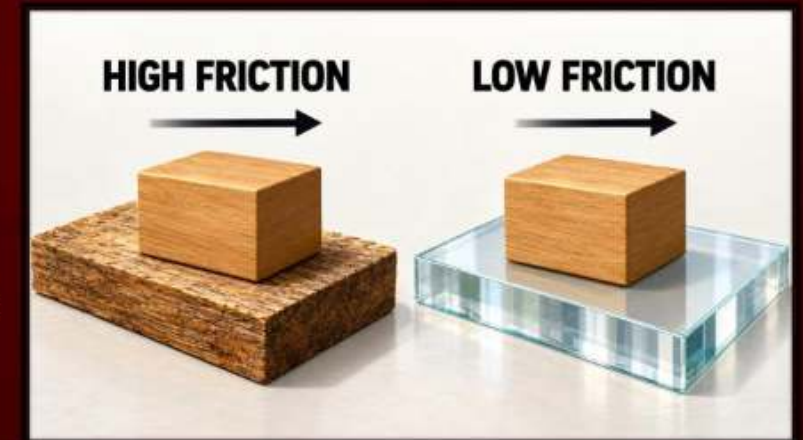




# Friction

## Method to reduce Friction

- ❖ Make the surfaces little more smoother.
- ❖ Lubrication is another way to make a surface smoother.
- ❖ Make the object more streamlined.



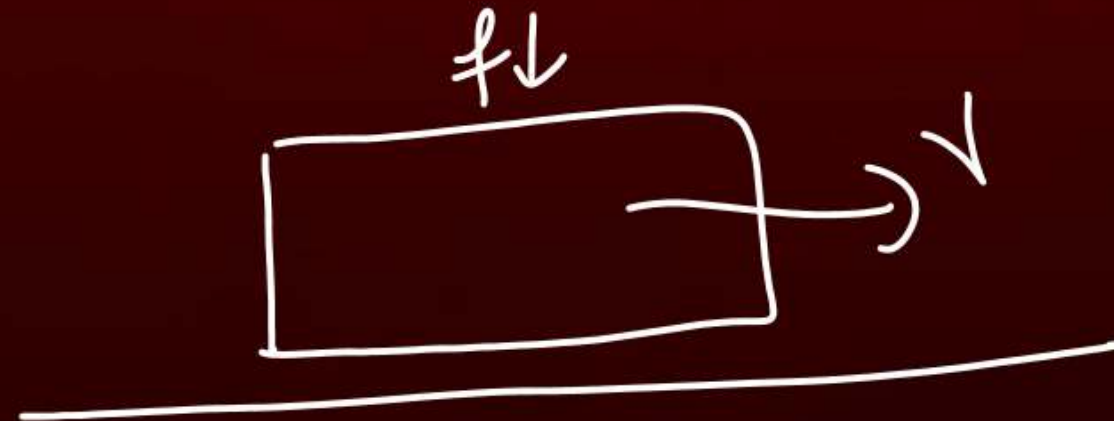


# Friction

$$\downarrow f = \mu N \downarrow$$

## Method to reduce Friction

- ❖ Reduce the Normal force acting between the surfaces in contact.
- ❖ Reduce the contact between the surfaces, so that less number of bonds will be formed.



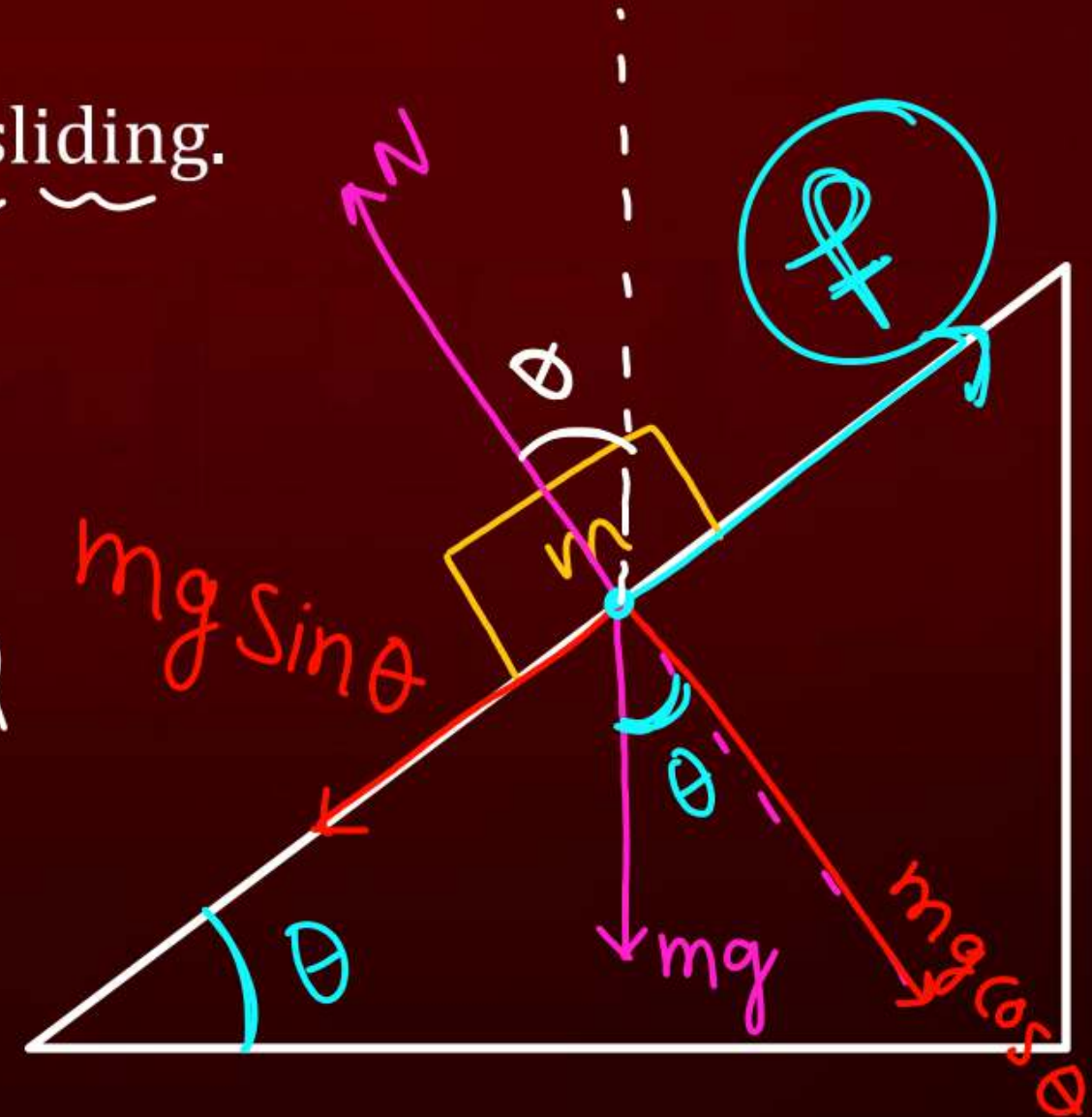


## Angle of Repose

The minimum angle when the body just starts sliding.

$$\begin{aligned} N &= mg \cos \theta \\ \text{Just slide} \\ mg \sin \theta &\approx f \\ mg \sin \theta &= \mu N \\ \cancel{mg} \sin \theta &= \mu \cancel{mg} \cos \theta \\ \frac{\sin \theta}{\cos \theta} &= \mu \end{aligned}$$

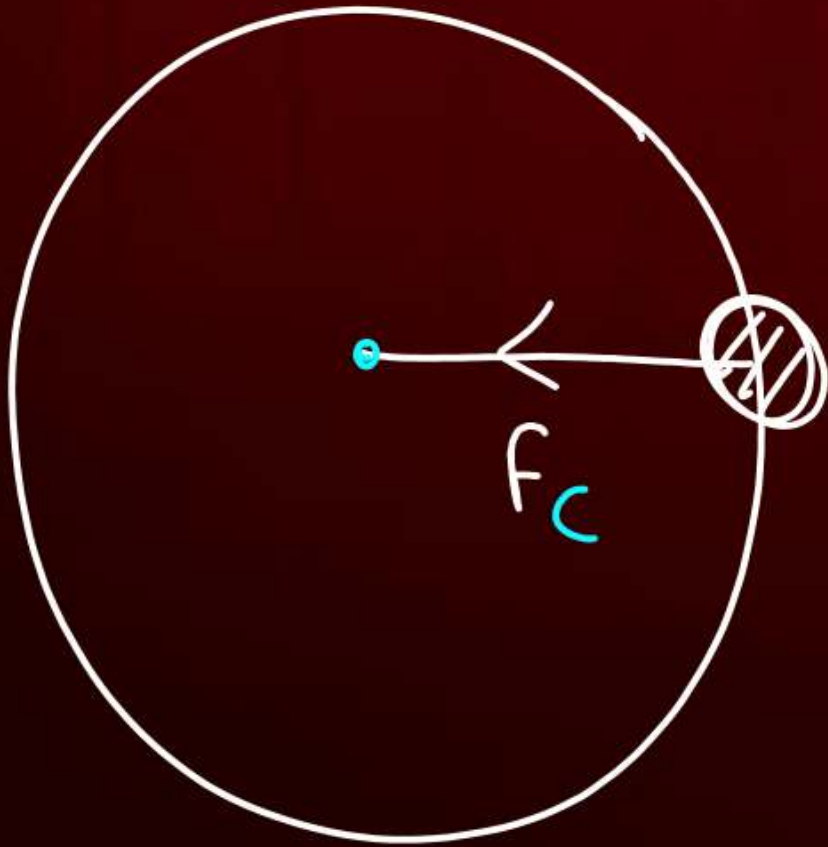
$$\begin{aligned} \frac{\sin \theta}{\cos \theta} &= \mu \\ \tan \theta &= \mu \end{aligned}$$





## Centripetal Force

Centripetal force is the force on an object on a circular path that keeps the object moving on the path. It is always directed towards the center.



$$F = ma$$

$$f_c = ma_c$$

$$f_c = m \frac{v^2}{R}$$

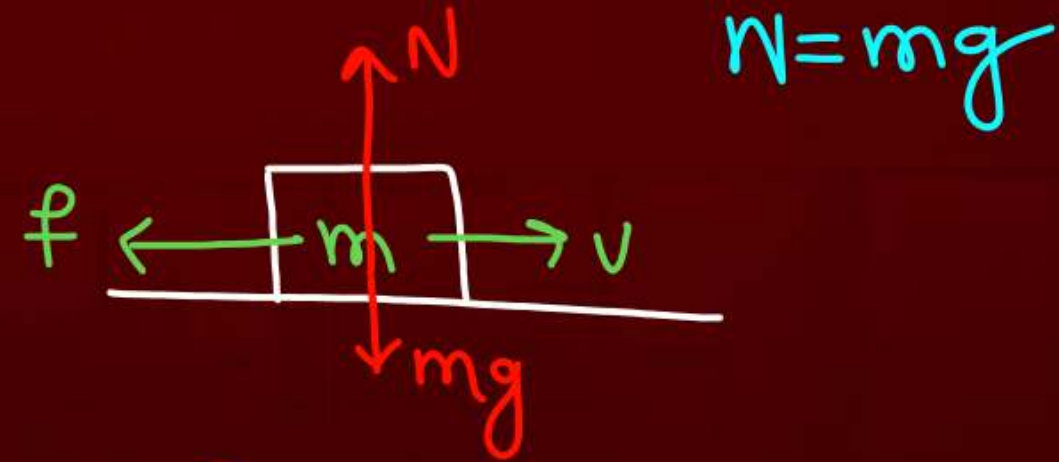
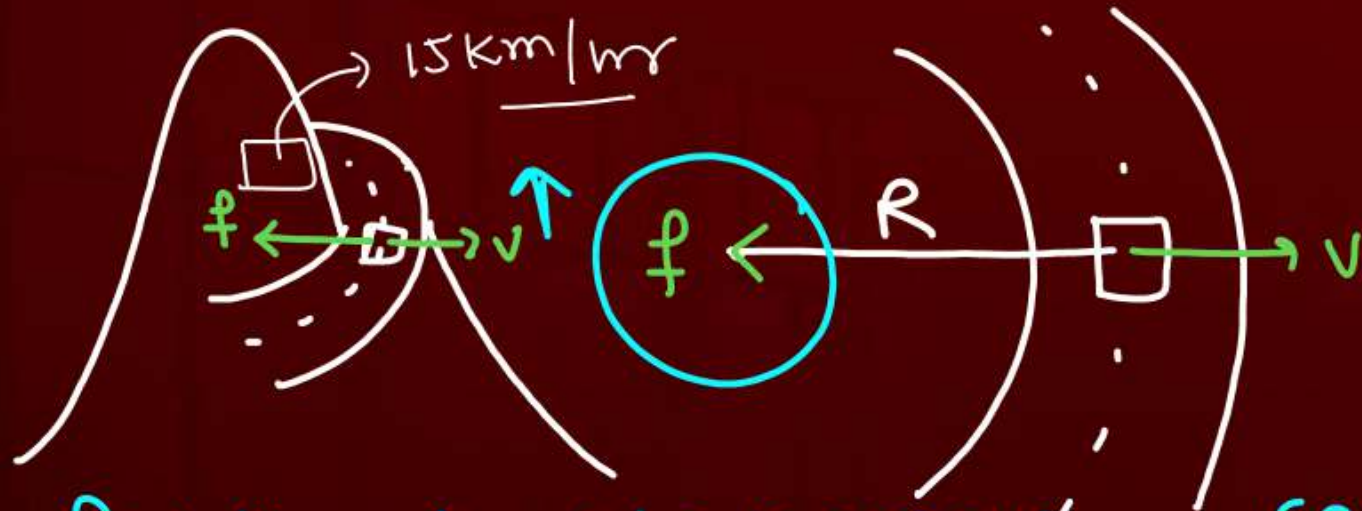
$$a_c = \frac{v^2}{R}$$





## a Maximum safe velocity on level road (during turning)

(i) when we make a turn on mountain (level Road)



$$N = mg$$

# friction provide necessary centripetal force

$$f = f_c$$

$$\mu N = \frac{mv^2}{R}$$

$$\cancel{\mu} \cancel{m} / \cancel{g} = \cancel{m} \cancel{v}^2 / \cancel{R}$$

$$\mu g = \frac{v^2}{R}$$

$$R\mu g = v^2$$

$$v^2 = R\mu g$$

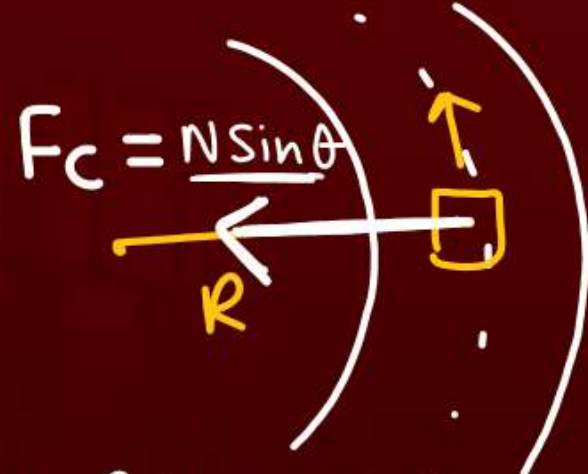
$$v^2 = \mu R g$$

$$V = \sqrt{\mu R g} \quad * * *$$

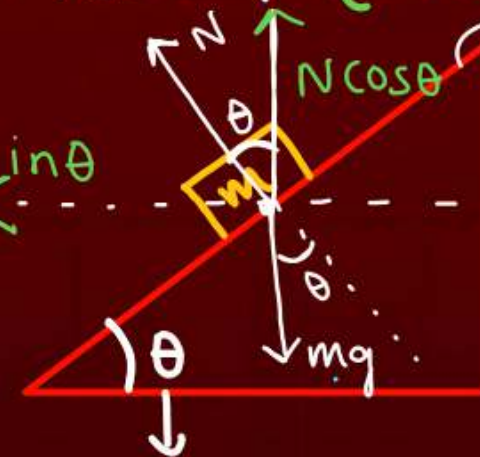


## (b) Maximum safe velocity on banked road

When you make a turn on a Banked Road (no-friction)



$$F_c = N \sin \theta$$



Angle of Banking

$$\mu = 0$$

$$f = \mu N$$

$$f = 0$$

the component of  $N$  ( $N \sin \theta$  provide) the necessary  $F_c$

$$N \sin \theta = F_c$$

$$N \sin \theta = \frac{mv^2}{R} \quad \text{--- (I)}$$

$$N \cos \theta = mg \quad \text{--- (II)}$$

Divide (I) and (II)

$$\frac{N \sin \theta}{N \cos \theta} = \frac{\frac{mv^2}{R}}{mg}$$

$$\tan \theta = \frac{v^2}{gR}$$

$$\tan \theta gR = v^2$$

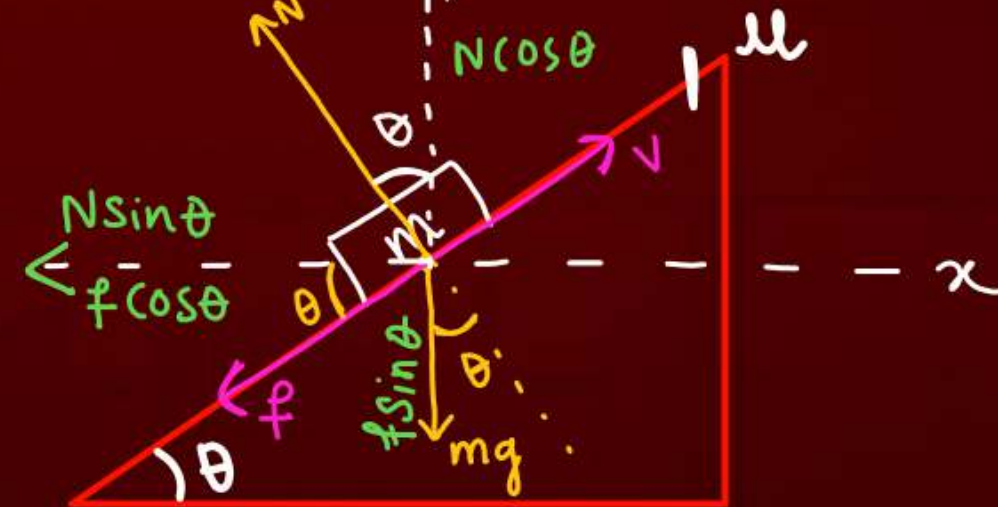
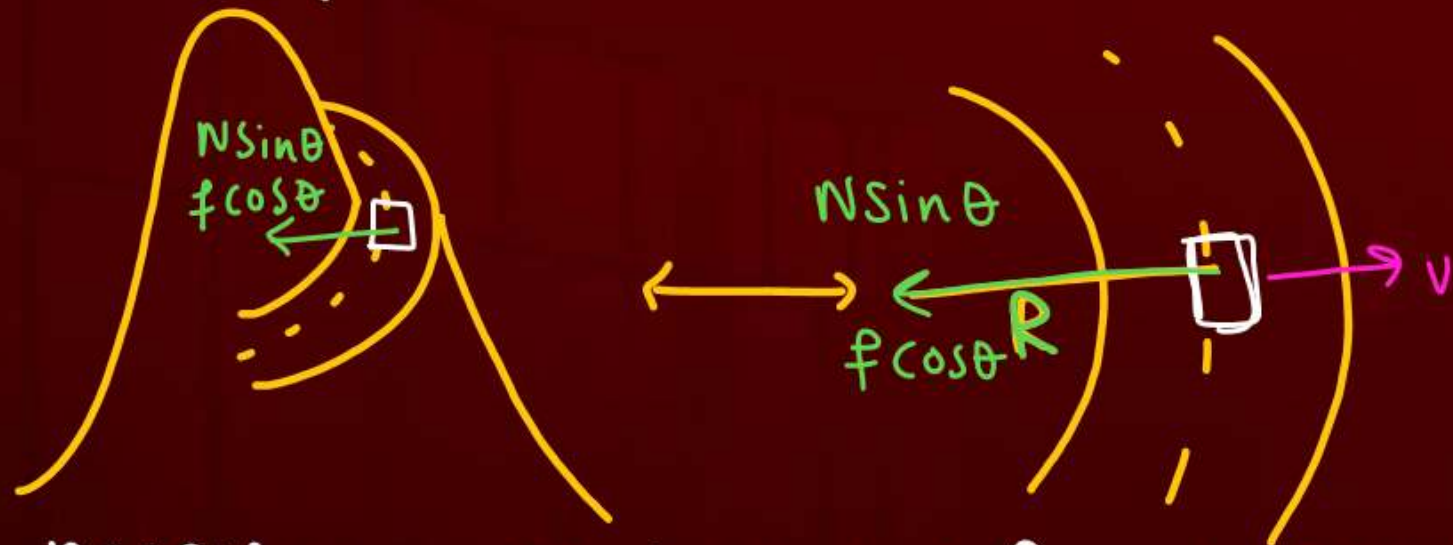
$$\sqrt{\tan \theta gR} = v$$

$$v = \sqrt{Rg \tan \theta}$$



## (c) Maximum safe velocity on banked road + Frictional Road

- when you take a turn on banked ( $\theta$ ) as well as frictional road



the necessary centripetal force is provided by  $N \sin \theta$  and  $f \cos \theta$

- $N \sin \theta + f \cos \theta = F_c$

$$N \sin \theta + f \cos \theta = \frac{mv^2}{R} \quad \text{--- (I)}$$

$$N \cos \theta = mg + f \sin \theta$$

$$N \cos \theta - f \sin \theta = mg \quad \text{--- (II)}$$

$$\frac{N \sin \theta + f \cos \theta}{\phantom{N \cos \theta - f \sin \theta}} = \frac{mv^2}{R} \quad \text{--- (1)}$$

$$N \cos \theta - f \sin \theta = mg$$

$$\frac{N \sin \theta + \overset{\text{f} = \mu N}{f \cos \theta}}{N \cos \theta - \underset{\text{f} = \mu N}{f \sin \theta}} = \frac{mv^2}{mgR}$$

$$\frac{N \sin \theta + \mu N \cos \theta}{\phantom{N \cos \theta - \mu N \sin \theta}} = \frac{v^2}{Rg} \quad \text{f} = \mu N$$

$$N \cos \theta - \mu N \sin \theta$$

Divide up and down by  $\cos \theta$

$$\frac{N \sin \theta + \mu N \cancel{\cos \theta}}{\cancel{N \cos \theta} - \mu N \sin \theta} = \frac{v^2}{Rg}$$

$$\frac{\cancel{N \cos \theta}}{\cancel{\cos \theta}} - \mu N \frac{\sin \theta}{\cos \theta}$$

$$\frac{N \tan \theta + \mu N}{N - N \mu \tan \theta} = \frac{v^2}{Rg}$$

$$\frac{\cancel{N} (\tan \theta + \mu)}{\cancel{N} (1 - \mu \tan \theta)} = \frac{v^2}{Rg}$$

$$\left[ \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right] = \frac{v^2}{Rg}$$

$$Rg \left[ \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right] = v^2$$

$$\sqrt{Rg \left[ \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right]} = v$$

$$V = \sqrt{Rg \left( \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)}$$

$$V = \sqrt{Rg \left( \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)}$$

only Banking

$\tan \theta \checkmark$

$\mu = 0$  (no friction)

$\mu$  (only friction)  
 $\tan \theta = 0$

$$V = \sqrt{Rg \left( \frac{\cancel{\tan \theta} + \mu}{1 - \mu \cancel{\tan \theta}} \right)}$$

$$V = \sqrt{Rg \frac{0 + \mu}{1 - \mu \times 0}}$$

$$V = \sqrt{\frac{Rg\mu}{1}}$$

$$V = \sqrt{\mu Rg}$$

$$V = \sqrt{Rg \left( \frac{\tan \theta + \cancel{\mu}}{1 - \cancel{\mu} \tan \theta} \right)^0}$$

$$V = \sqrt{Rg \left( \frac{\tan \theta + 0}{1 - 0} \right)}$$

$$V = \sqrt{Rg \tan \theta}$$



## Question

2023

**Define angle of repose. Show that coefficient of friction is equal to the tangent of angle of repose.**



## Question

5m  
2024

**Define centripetal force. Derive expression for it.**



## Question

h/w  
**State Newton's three law of motion. Show that Newton's second law is the only real law of motion.**



## Question

**State Newton's three laws of motion.**

2024



## Question

**Explain the term impulse. Give one example.**

3m

2021

# Angle of friction

The Angle b/w the Resultant of (f) and (N) with Normal

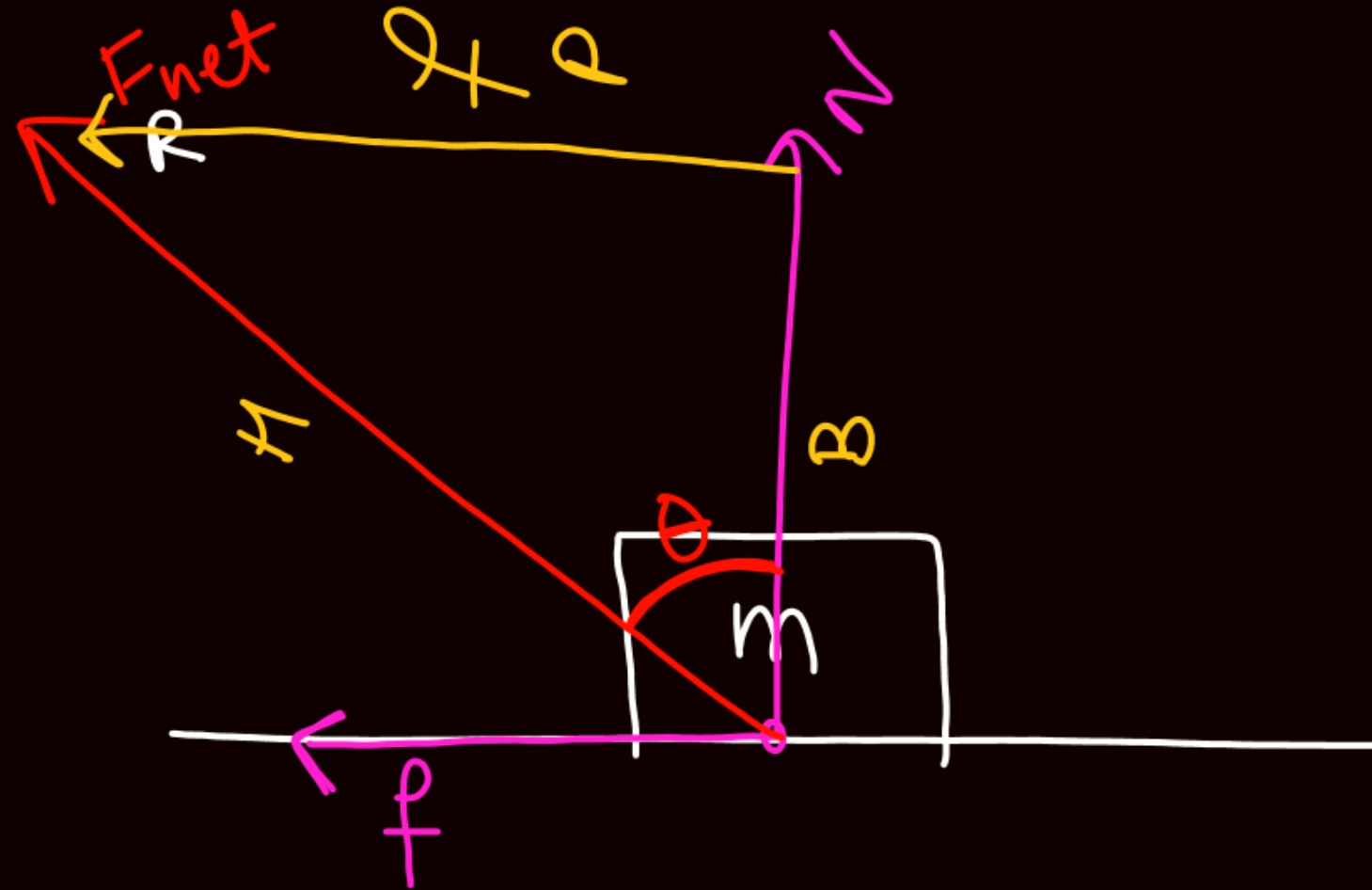
$$\tan \theta = \frac{P}{B}$$

$$\tan \theta = \frac{f}{N}$$

$$\tan \theta = \frac{N}{\cancel{N}}$$

$$f = \mu N$$

$$\tan \theta = \mu$$



**Thank  
You**